

Cybersecurity Foundation & Environment Setup

Task 1 – Lab Setup Report

1. Introduction

Cybersecurity requires practical knowledge of operating systems, networking, security tools, and common attack techniques. This lab is designed to provide a safe and controlled environment for learning and practicing basic cybersecurity concepts.

In this task, a virtual cybersecurity lab is created using **Kali Linux** as the testing machine and **Metasploitable 2** or **DVWA** as the vulnerable target. The lab also covers Linux fundamentals, networking, cryptography, and basic security tools such as Nmap, Wireshark, Burp Suite, and Netcat.

2. Objectives

The main objectives of this lab are:

- To understand the fundamentals of cybersecurity.
- To learn the CIA Triad — Confidentiality, Integrity, and Availability.
- To understand common cybersecurity threats and attack vectors.
- To set up a virtual cybersecurity laboratory.
- To learn basic Linux commands and file permissions.
- To understand basic networking concepts.
- To practice basic cryptography concepts.
- To become familiar with Nmap, Wireshark, Burp Suite, and Netcat.
- To perform security testing in an isolated and controlled environment.

3. Cybersecurity Basics

3.1 CIA Triad

CIA Triad

The CIA Triad is a fundamental security model consisting of:

Confidentiality: Ensures that information is accessible only to authorized users.

Integrity: Ensures that information is accurate and has not been modified without authorization.

Availability: Ensures that systems and information are available when required by authorized users.

3.2 Common Threats

Some common cybersecurity threats studied in this task include:

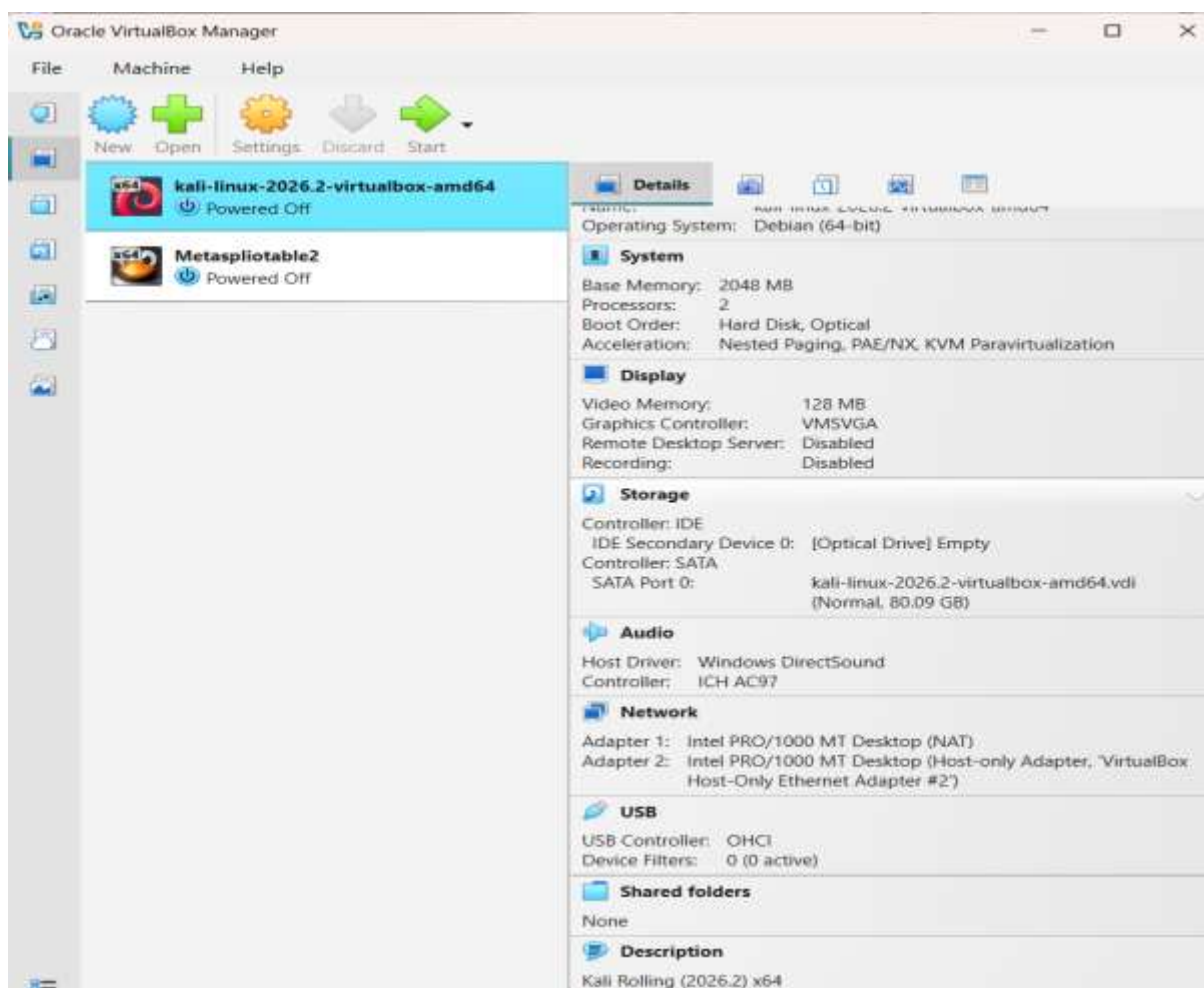
- **Phishing** – Attempting to trick users into revealing sensitive information.
- **Malware** – Malicious software designed to damage, disrupt, or gain unauthorized access.
- **DDoS** – Overwhelming a service with large amounts of traffic.
- **SQL Injection** – Exploiting improperly handled database queries.
- **Brute Force** – Trying many possible passwords or credentials.
- **Ransomware** – Malware that restricts access to data or systems and demands payment.

4. Lab Environment Setup

4.1 Virtualization Software: A virtualization platform such as **VirtualBox** or **VMware** used to create and run virtual machines.

Software Used: VirtualBox

Purpose: To create an isolated environment for cybersecurity practice.



4.2 Kali Linux Setup

Kali Linux is used as the **attacker/testing machine**. It provides many pre-installed tools useful for cybersecurity learning and penetration testing.

1. Install Kali Linux as a virtual machine.
2. Start the Kali Linux VM.
3. Log in to the system.
4. Open the terminal.
5. Check the network configuration.

```
(kali@kali)~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group def
link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
inet 192.168.128.3/24 brd 192.168.128.255 scope global dynamic noprefixroute eth0
    valid_lft 82486sec preferred_lft 82486sec
inet6 fd17:625c:f937:2:a9f3:6307:3cf8:3f0/64 scope global dynamic noprefixroute
    valid_lft 86368sec preferred_lft 14368sec
inet6 fe80::9a4f:830f:9bc8:a0a3/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group def
link/ether 08:00:27:b7:ed:d9 brd ff:ff:ff:ff:ff:ff
inet 192.168.128.4/24 brd 192.168.128.255 scope global dynamic noprefixroute eth1
    valid_lft 384sec preferred_lft 384sec
inet6 fe80::cec0:794f:98ac:287f/64 scope link noprefixroute
    valid_lft forever preferred_lft forever

(kali@kali)~$ ping -c 4 192.168.128.3
PING 192.168.128.3 (192.168.128.3) 56(84) bytes of data:
64 bytes from 192.168.128.3: icmp_seq=1 ttl=64 time=1.70 ms
64 bytes from 192.168.128.3: icmp_seq=2 ttl=64 time=1.40 ms
64 bytes from 192.168.128.3: icmp_seq=3 ttl=64 time=1.10 ms
64 bytes from 192.168.128.3: icmp_seq=4 ttl=64 time=1.41 ms

--- 192.168.128.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.099/1.402/1.703/0.213 ms

(kali@kali)~$
```

5. Metasploitable 2 / DVWA Setup

Metasploitable 2 is an intentionally vulnerable virtual machine designed for security training.

It is used as the **target machine** in this lab. The target machine should be connected only to the isolated lab network

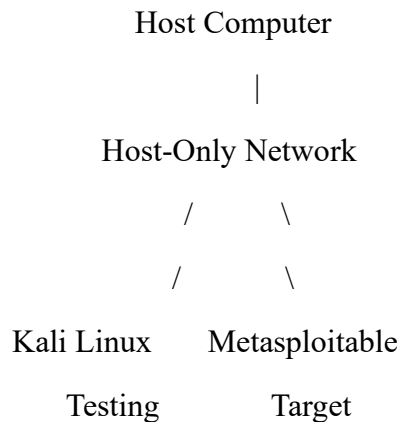
```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:f2:6e:2d
          inet addr:192.168.128.3  Bcast:192.168.128.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fef2:6e2d/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:13 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3492 (3.4 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$
```

6. Network Configuration

A **Host-Only Adapter** can be used to create an isolated network between the virtual machines.



The purpose of this configuration is to allow the virtual machines to communicate with each other without exposing the vulnerable target unnecessarily to the external network.

Connectivity Test

From Kali Linux:

```
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 82486sec preferred_lft 82486sec
    inet6 fd17:625c:f037:2:69f3:6302:3cfb:3f0/64 scope global dynamic noprefixroute
        valid_lft 86368sec preferred_lft 14368sec
    inet6 fe80::9e4f:630f:9bc8:40a3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 08:00:27:b7:ed:d9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.128.4/24 brd 192.168.128.255 scope global dynamic noprefixroute eth1
        valid_lft 384sec preferred_lft 384sec
    inet6 fe80::cec0:794f:90ec:287f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping -c 4 192.168.128.3
PING 192.168.128.3 (192.168.128.3) 56(84) bytes of data.
64 bytes from 192.168.128.3: icmp_seq=1 ttl=64 time=1.70 ms
64 bytes from 192.168.128.3: icmp_seq=2 ttl=64 time=1.40 ms
64 bytes from 192.168.128.3: icmp_seq=3 ttl=64 time=1.10 ms
64 bytes from 192.168.128.3: icmp_seq=4 ttl=64 time=1.41 ms

--- 192.168.128.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.099/1.402/1.703/0.213 ms

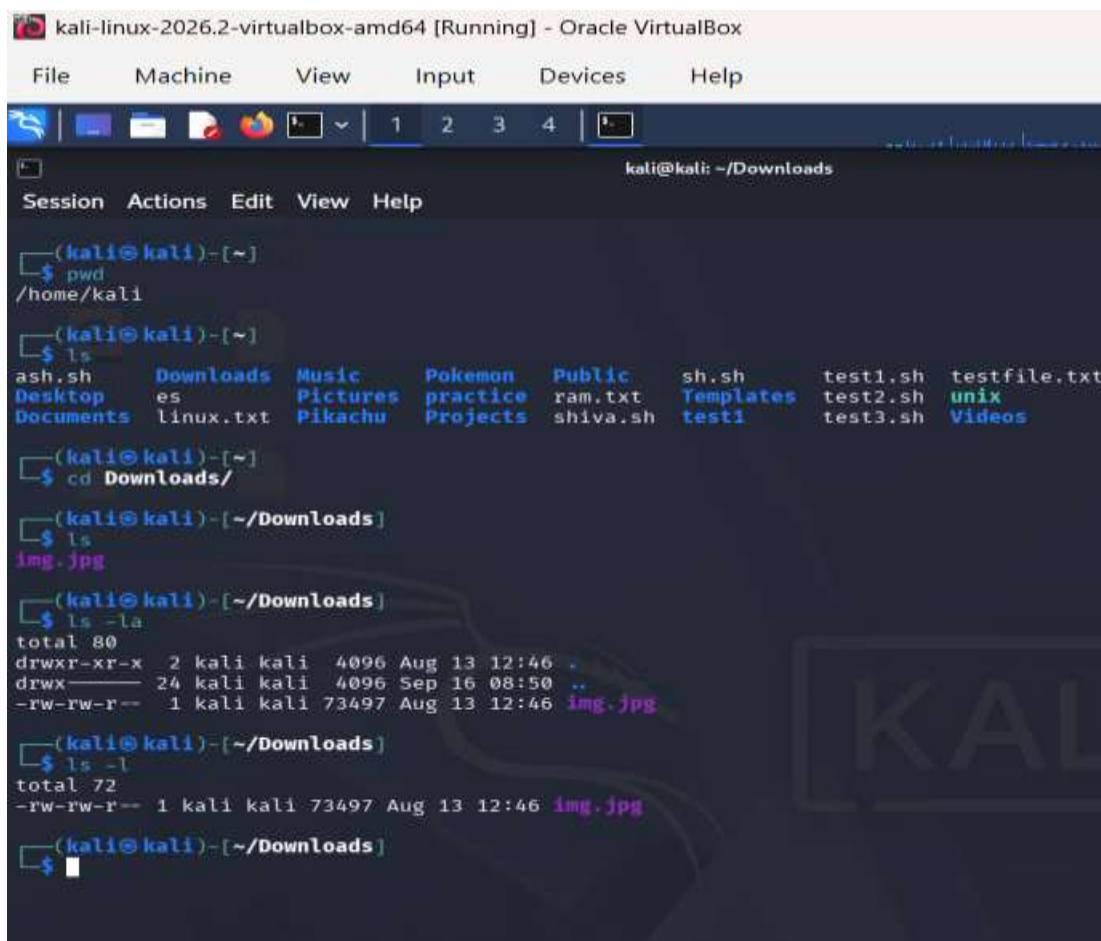
(kali@kali)-[~]
$
```

7. Linux Fundamentals

During the lab, basic Linux commands were practiced.

Command Purpose

pwd	Displays the current directory
ls	Lists files and directories
cd	Changes directory
mkdir	Creates a directory
cp	Copies files
mv	Moves/renames files
rm	Removes files
chmod	Changes file permissions
chown	Changes file ownership



The screenshot shows a Kali Linux terminal window titled "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox". The terminal displays the following commands and output:

```
(kali@kali)-[~]
$ pwd
/home/kali

(kali@kali)-[~]
$ ls
ash.sh  Downloads  Music  Pokemon  Public  sh.sh  test1.sh  testfile.txt
Desktop es         Pictures practice ram.txt  Templates test2.sh  unix
Documents linux.txt  Pikachu Projects shiva.sh test3.sh  Videos

(kali@kali)-[~]
$ cd Downloads/

(kali@kali)-[~/Downloads]
$ ls
img.jpg

(kali@kali)-[~/Downloads]
$ ls -la
total 80
drwxr-xr-x  2 kali kali  4096 Aug 13 12:46 .
drwx----- 24 kali kali  4096 Sep 16 08:50 ..
-rw-rw-r--  1 kali kali 73497 Aug 13 12:46 img.jpg

(kali@kali)-[~/Downloads]
$ ls -l
total 72
-rw-rw-r--  1 kali kali 73497 Aug 13 12:46 img.jpg
```

8. Networking Fundamentals

The following networking concepts were studied:

```
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 82486sec preferred_lft 82486sec
    inet6 fd17:625c:f037:2:69f3:6302:3cfb:3f0/64 scope global dynamic noprefixroute
        valid_lft 86368sec preferred_lft 14368sec
    inet6 fe80::9e4f:630f:9bc8:40a3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b7:ed:d9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.128.4/24 brd 192.168.128.255 scope global dynamic noprefixroute eth1
        valid_lft 384sec preferred_lft 384sec
    inet6 fe80::cec0:794f:90ec:287f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping -c 4 192.168.128.3
PING 192.168.128.3 (192.168.128.3) 56(84) bytes of data.
64 bytes from 192.168.128.3: icmp_seq=1 ttl=64 time=1.70 ms
64 bytes from 192.168.128.3: icmp_seq=2 ttl=64 time=1.40 ms
64 bytes from 192.168.128.3: icmp_seq=3 ttl=64 time=1.10 ms
64 bytes from 192.168.128.3: icmp_seq=4 ttl=64 time=1.41 ms

--- 192.168.128.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.099/1.402/1.703/0.213 ms

(kali@kali)-[~]
$
```

9. Cryptography Basics

Cryptography is used to protect information from unauthorized access.

Symmetric Encryption : Uses the same key for encryption and decryption.

Asymmetric Encryption : Uses a public key and a private key.

Hashing

Hash functions convert data into a fixed-length value. Examples include:

- MD5
- SHA-256

SSL/TLS

SSL/TLS protocols help provide secure communication over networks, particularly for HTTPS.

OpenSSL

Basic encryption and decryption operations can be explored using OpenSSL.

10. Security Tools

10.1 Nmap

Nmap is a network scanning tool used to discover hosts and services on a network.

Example: nmap <target-IP>

```
(kali@kali)-[~]
$ nmap -sV 192.168.128.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 05:23 -0400
Nmap scan report for 192.168.128.3
Host is up (0.00061s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:F2:6E:2D (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN;
_kernel

Service detection performed. Please report any incorrect results at https
Nmap done: 1 IP address (1 host up) scanned in 14.27 seconds

(kali@kali)-[~]
$ nmap -p 21 -sV 192.168.128.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 05:25 -0400
Nmap scan report for 192.168.128.3
Host is up (0.0011s latency).

PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
MAC Address: 08:00:27:F2:6E:2D (Oracle VirtualBox virtual NIC)
Service Info: OS: Unix
```

10.2 Wireshark

Wireshark is a network protocol analyzer used to capture and inspect network packets.

Basic process:

1. Open Wireshark.
2. Select the appropriate network interface.
3. Start packet capture.
4. Generate some traffic, such as a ping.
5. Stop the capture.
6. Examine the packets.

10.3 Burp Suite

Burp Suite is a web security testing platform. Its proxy functionality allows HTTP/HTTPS requests and responses to be inspected during authorized testing.

Conclusion

This task provided a foundation in cybersecurity by combining theoretical concepts with practical laboratory activities. The lab environment enabled the exploration of Linux, networking, cryptography, and security tools in a controlled environment. The knowledge gained from this task provides a foundation for further cybersecurity learning and practical exercises.